

JAD FALAQ

Data & AI Engineer – Data Pipelines, Machine Learning & Generative AI

Driving licence B | Age 22 | jadfalaq@gmail.com
linkedin.com/in/jadfalaq | github.com/JadFalaq | www.jadfalaq.online

6-month final-year engineering internship starting January 2027

PROFILE

Data & AI Engineer, final-year AI engineering student at **ENSIAS**, with internships at **Renault Digital** and **AKWA Group**. Builds **end-to-end data pipelines**, from ingestion to production, including **anomaly detection on 6M+ transactions** integrated into the existing system, a **document ingestion pipeline** feeding a generative AI system, and **image, video and text data processing**. Hybrid **Data Engineering / AI** profile focused on **data quality**.

PROFESSIONAL EXPERIENCE

AI Engineer Intern – Quality Engineering

Renault Digital | Internship | Casablanca, Morocco Jun. 2026 – Sep. 2026

- **Document ingestion pipeline & RAG.** Ingestion of documentation and past tickets (OCR of tables and images with unstructured.io, document-structure chunking, embeddings in pgvector, MMR retrieval) feeding a multi-agent AI assistant that drafts Jira tickets, created only after human validation, with sensitive data masked. Writing time cut from 10–20 min to 1–2 min per ticket.
- **Workflow automation.** Four Power Automate workflows launched on my own initiative, generating the test-tracking files (Smoke, TNR), reminding about empty statuses and handling error cases (no duplicates, team lead alerted), in Agile sprints.

ML & Data Engineer – IT Data Team

AKWA Group | Internship | Casablanca, Morocco Jun. 2025 – Sep. 2025

- **Data engineering & ERP integration.** Reconciliation of accounting entries from Afriware, the Odoo and JDE ERPs and internal files across 6M+ transactions, then automated integration into the existing system.
- **End-to-end ML pipeline (CRISP-DM).** Preprocessing, feature engineering and comparison of supervised and unsupervised models (Isolation Forest, autoencoder). Random Forest selected (98% accuracy, 85% F1-score).

TECHNICAL SKILLS

- **Data processing & preprocessing.** Pandas, NumPy, SciPy, scikit-learn, OpenCV, torchvision, librosa, spaCy, pypdf, data cleaning, deduplication, data quality, EDA, feature extraction, up/downsampling, data leakage prevention
- **Dimensionality reduction.** PCA, LDA, t-SNE, latent space (autoencoders)
- **Data engineering & orchestration.** SQL, ETL, Spark, Airflow, BigQuery, JSON/CSV/XLSX, Pydantic, DVC, CRISP-DM
- **Databases.** PostgreSQL, MongoDB, Neo4j (Cypher), SQLite, SQLAlchemy, pgvector, Chroma
- **BI & visualization.** Power BI, Looker, Tableau, Streamlit, Plotly, Matplotlib, Seaborn
- **ML & deep learning.** XGBoost, PyTorch, TensorFlow, YOLOv8, DeepSORT, LSTM, encoder-decoder, knowledge distillation
- **Generative AI & NLP.** RAG, Azure OpenAI, Microsoft Copilot, LangChain, LangGraph, BM25, MMR, RRF
- **Cloud, DevOps & back-end.** Docker, Git, GitHub Actions (CI/CD), Azure Container Apps, MLflow, Python, FastAPI, Power Automate

EDUCATION

State Engineering Degree – Artificial Intelligence Engineering (2IA)

ENSIAS, Rabat | Sep. 2024 – Jun. 2027

- Machine learning and deep learning (neural networks, reinforcement learning), statistical and probabilistic learning
- Big Data and distributed systems, data engineering, time series, multimedia processing (text, image)

Preparatory Classes for the French Grandes Écoles (CPGE) – Mathematics & Physics

CPGE Mohammed V, Casablanca | Sep. 2022 – Jun. 2024

ACADEMIC PROJECTS

- **Dental pathology detection – Data & MLOps.** Audit and cleaning of 1,808 panoramic X-rays (24,891 annotations, duplicates, invalid labels), remapping into 7 clinical classes and a reproducible DVC pipeline. YOLOv8s training, 3 variants compared on an external split and tracked in MLflow, champion selected for its recall, Dockerized FastAPI service (image reduced from 3.74 to 1.19 GB) and CI/CD
- **AmnAI – Video sequence analysis.** Real-time pipeline on surveillance videos with person detection (YOLO), tracking (DeepSORT), pose (MediaPipe) and behavioral features over time windows, classified by a BiLSTM with attention, 80–92% precision at 20–28 FPS
- **cvmatch – CV processing & hybrid search.** Ingestion of CVs and job offers, skill extraction, evidence retrieval with BM25 + embeddings (RRF fusion, FR/EN synonyms), parallel skill assessment (LangGraph), coverage map, 3 scores and rewriting suggestions

SOFT SKILLS

- **Entrepreneurial mindset.** Automations launched on my own initiative, e-commerce websites built for clients
- **Listening & ownership.** Product shaped by user feedback, no AI action without human validation

LANGUAGES

French (native) – Arabic (native) – English (bilingual)